

THE EMERGENT OF INTERFACES AND INTERACTIONS



CHAPTER Ten

Instructor Name :- Suzana Binti Ahmad

? Interface

- Medium of communication between objects and person

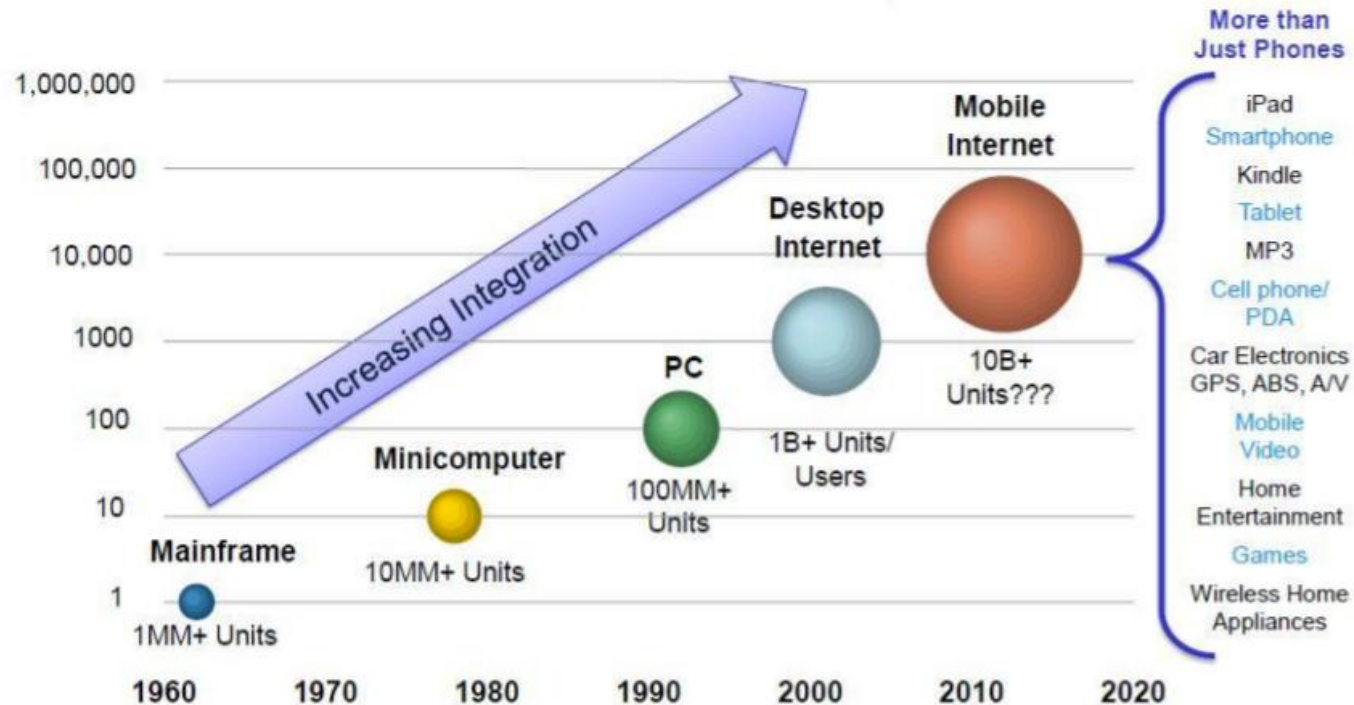


Emergence of interface

- Initial developments
 - Input devices were used for direct control of computer-created data
 - Later.. Cathode-ray-tube (CRT) screens used a lightpen, rather than a mouse, as the pointing device
 - Concept of a multi-panel windowing system



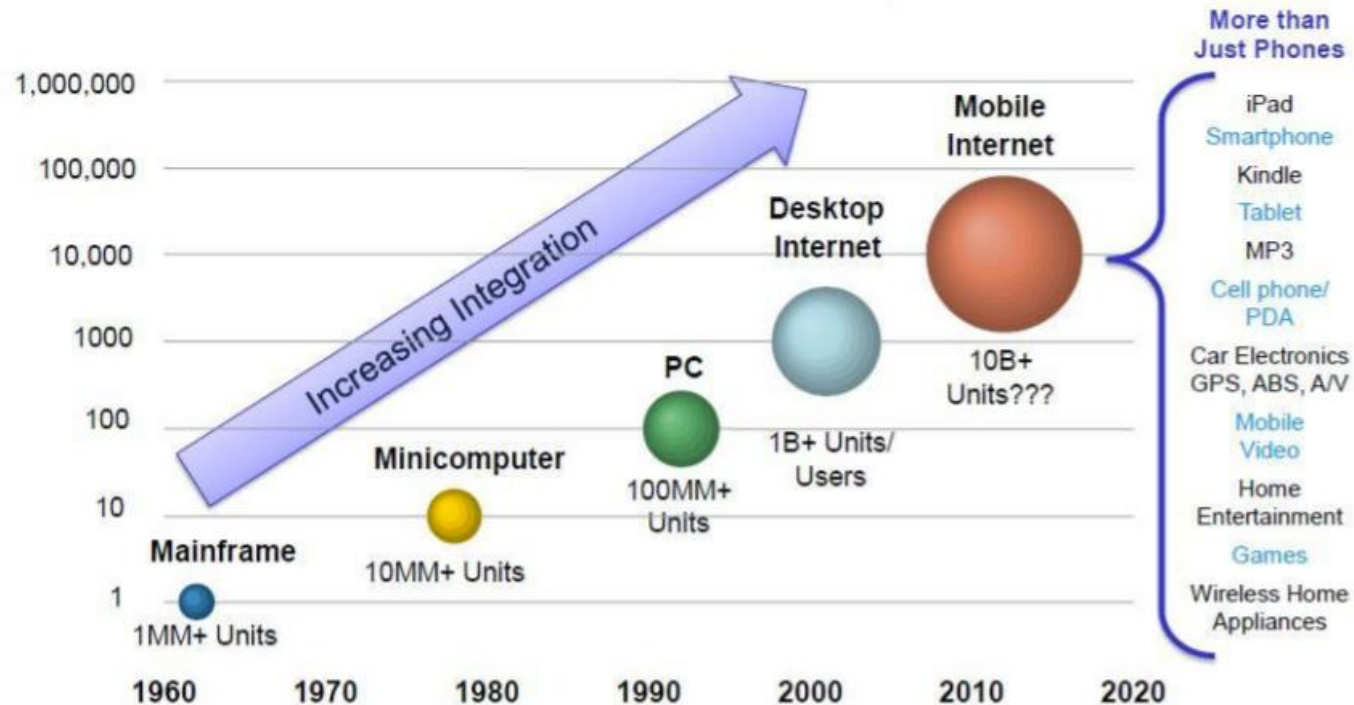
Emergence of interface



Note: PC installed base reached 100MM in 1993, cellphone/internet users reached 1B in 2002/2005 respectively;
Source: ITU, Mark Lipacis, Morgan Stanley Research.

- With the current ability of 'connection' regardless of location - generated new avenues (access content, consumer behaviour).
- user interface and interaction evolve to match consumer expectations of more seamless, intuitive and immersive.

Emergence of interface



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- over the years size and number of units sold, improvement degree of interactivity and degree of integration with other devices and technologies.
- evolution marks the beginning of a transformational shift - user experience through computing form factors superseding the effects of product

Market Trends

3 main drivers for user interface and interaction technologies

- visual info-gratification
- the user as the new interface
- smarter devices and emerging user technologies

Market Trends

Visual info-gratification

- consumers accessing information about real-world objects naturally and on-the-go
- as simply by pointing their smart devices at an object of interest

Market Trends

The user as the new interface

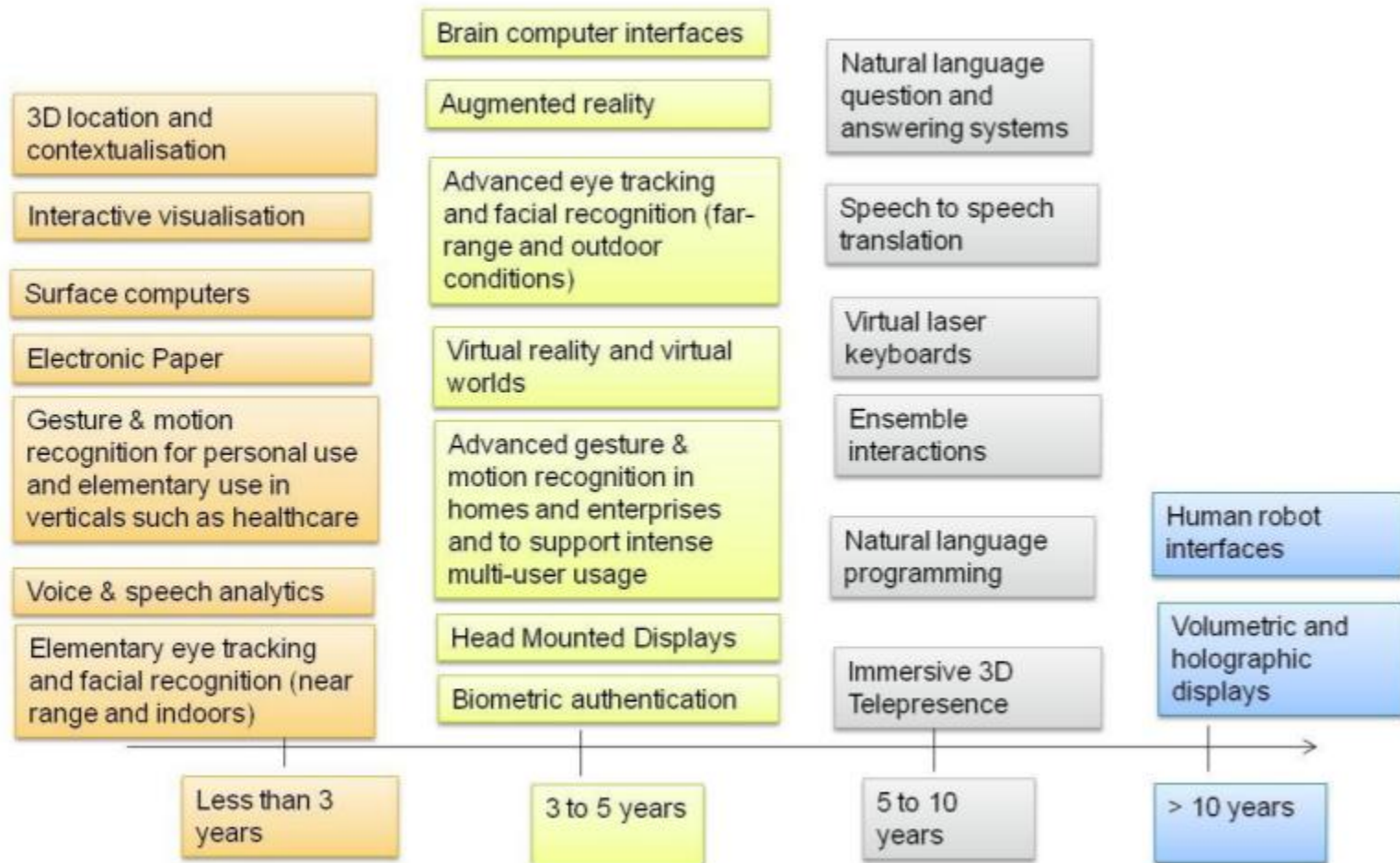
- the concept of natural user interface shattered the existing paradigm of user interfaces by positioning users as the new user interface
- Microsoft Kinect was developed - able to decipher images, gestures and speech commands from users

Market Trends

Smarter devices and emerging user technologies

- Multiple user interface technologies for immersive environments (extends for industrial to consumer context)
- technologies are evolving to enhance the user experience, eg. surface computers (computers interact with users through the surface of ordinary objects rather than through monitor and keyboard)
- more sophisticated forms of media and enhanced experiences

Technology Outlooks



Timeline of enabling technologies

Technology Outlooks

- Electronic (e-)paper - reflective display technologies that do not require a backlight; can be viewed in conditions of moderate to good ambient illumination.



Technology Outlooks

- Interactive visualization - displays data using interactive images with the colour brightness, size, shape and motion of visual objects representing aspects of the dataset being analyzed.
- these products provide an array of visualization options, eg. heat maps, geographic maps, scatter plots and other special-purpose visuals.

Technology Outlooks

- Seamless indoor and outdoor location positioning and contextualization.
- common location determination technologies provide location positioning

Technology Outlooks

- Surface Computers
- large-screen displays that support direct interaction via multi-touch or gesture
- enabling multiple users to interact or work collaboratively

Technology Outlooks

- Gesture Recognition and Motion Sensing for Elementary use
- involves determining the movement of a user's fingers, hands, arms, head or body in three dimensions through the use of a camera or device with embedded sensors that may be worn, held or body-mounted

Technology Outlooks

- Elementary Eye Tracking and Facial Recognition
- determine the angle or position of a user's visual attention, typically through the use of camera.

Technology Outlooks

- Augmented Reality
- Superimposes graphics, audio, other virtual enhancements over a live view or the real world
- Aims to enhance user interaction with the environment in real time through the digital environment

Technology Outlooks

- Advanced Gesture Recognition and Motion Detection
- Analysing 2/3 users gestures simultaneously, with gestural commands mainly used in gaming context

Technology Outlooks

- Advanced Eye Tracking and Facial Recognition
- Determine the facial expressions and eye gazes of multiple users from great distances at the same time and in dynamically changing environments such as different lighting conditions with moving crowds

Technology Outlooks

- Virtual reality and virtual worlds
- Provide computer-generated 3D environments that surround a user and respond naturally to his actions.

Technology Outlooks

- Wearable computers
- Designed to be worn on the body to enable immersive user experience, mobility and hands-free activities



Technology Outlooks

- Head-mounted displays
- Designed to be worn on the head to enable an immersive user experience, mobility and hands-free activities.

Technology Outlooks

- Brain computer interfaces
- Interpret distinct brain patterns, shifts and signals
- Non-invasive techniques use a cap or helmet to detect the signals through external electrodes.

Technology Outlooks

- Advanced natural language question and answering
- Provides users with a means of asking a question in plain language of a computer of service and receiving an answer of high reliability and confidence in a reasonable time frame.

Technology Outlooks

- Immersive 3D telepresence
- Provides a high-definition immersive experience, enabling the illusion of presence of geographically dispersed meeting participants in the meeting.

Technology Outlooks

- Ensemble interactions
- Enable seamless user experiences across multiple personal devices.
- Interactions will easily shift to the most appropriate device as and when required by the user

Technology Outlooks

- Haptics
- Use of tactile interfaces (vibration or pressure)
- Provide touch or force feedback as part of the user interface

Technology Outlooks

- Speech-to-speech translation
- Involves translating one spoken language into another.
- Combines speech recognition, machine translation and text-to-speech capabilities

Technology Outlooks

- Natural language programming
- Envisaged surge in tablet adoption and other smart mobile devices will lead users to develop software programmes and apps directly on these device.
- Given the limited screen size, information layout and difficulties in typing on virtual keyboards, these devices are not optimal in enabling users to create new software programmes and apps productively.

Technology Outlooks

- Virtual laser keyboards
- Small projector modules that displays user interface onto any flat surface so that users interact more easily with mobile devices or systems (without physical)



Technology Outlooks

- Human-Robot-Interface
- Convergence of 3D location and contextualization, natural language question and answering systems, speech-to-speech translation, gesture recognition, eye tracking and facial recognition – intelligent human-robot interface



Technology Outlooks

- Volumetric (beyond holographic) displays
- Create visual 3D representations of objects, with an almost 360-degree spherical viewing angle



Market Application

- Education
- Healthcare
- Government
- Entertainment
- Market intelligence
- Retail

Implications

- Consumers/Humanization
- Enterprises/Industries
- Government
- R&D

End of Lecture